Growth Accounting: Japan 1995 - 2015

Macroeconomics I

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March 2026

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A Country & Time Period

Identify a time period and a country of your interest for which you can get data.

This growth accounting exercise explores the experience of Japan between 1995 and 2015.

Motivation: Japan between 1995–2015 offers a compelling case study for a growth accounting exercise. Japan over this period is characterised by persistently modest growth, despite being one of the world's largest economies. It faced a varied set of structural and macroeconomic challenges, making it an interesting setting for a growth accounting exercise.

Note: It should be noted that growth accounting does not enable us to make any causal statements about the effect of policies on the economy. Additionally, we are aware that potential misallocation in the economy may distort our decomposition results. Specifically, the models we employ are based on the assumption that marginal products are equalised across firms. In reality, frictions likely prevent firms from choosing their optimal input mix. This distorts output as compared to its optimum level. This means that measured total factor productivity (TFP) growth - computed as a residual - may be biased, and the estimated contributions of capital and labour may be distorted.

B Output Per Worker

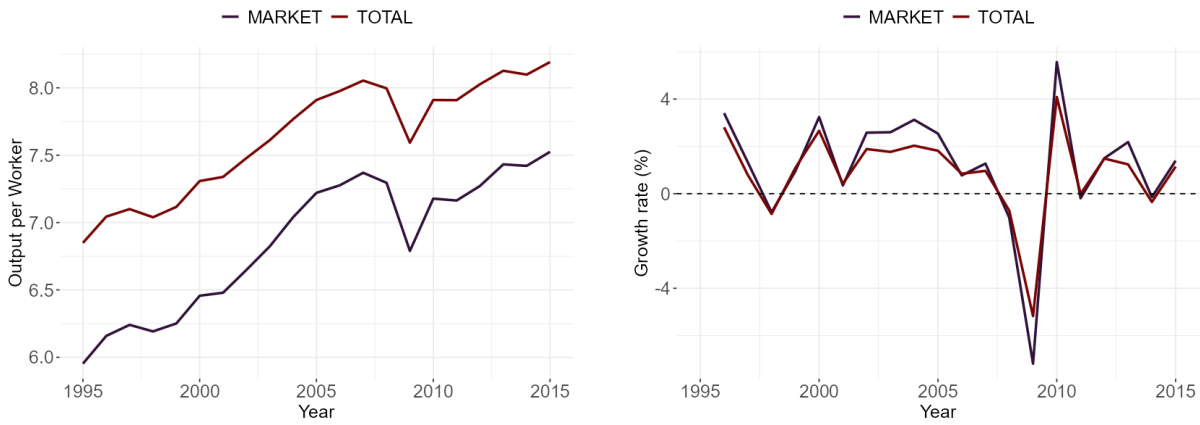
Describe the evolution of aggregate output per worker. Discuss major policies in place (or policy changes), institutional features etc that characterize the country and time period of your choosing. Does it matter whether you focus on the total of the economy or the market economy?

Data choices: To analyse output per worker, we analyse three different measurements of *worker* that are available in the EUKLEMS database:

1. **Number of Employees** [*EMP*]
2. **Total Hours Worked by Person Engaged** [*H_EMP*]
3. **Total Hours Worked by Employees** [*H_EMPE*]

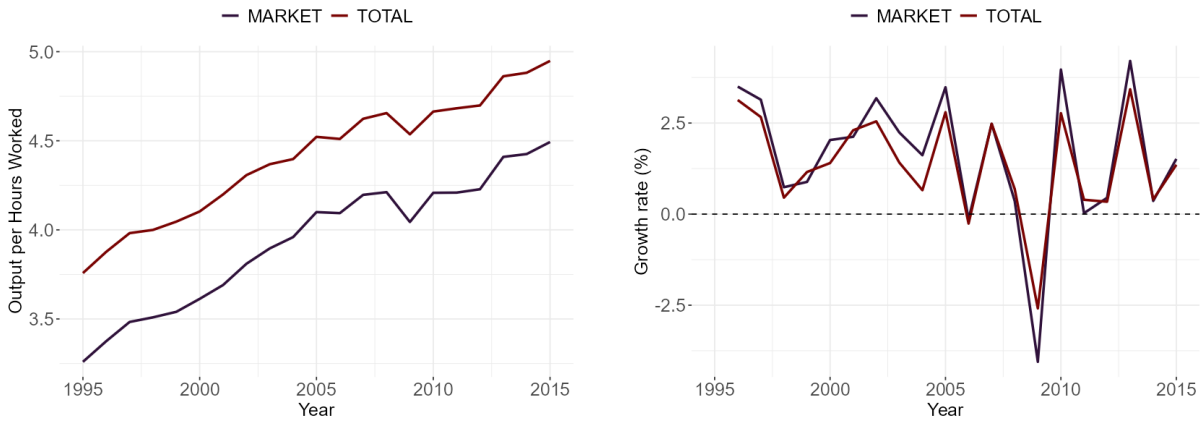
Figures

Figure 1, 2 and 3 plot output per worker for each of these three measures respectively, showing both the level and growth rate for the total and market economy in Japan between 1995 and 2015.



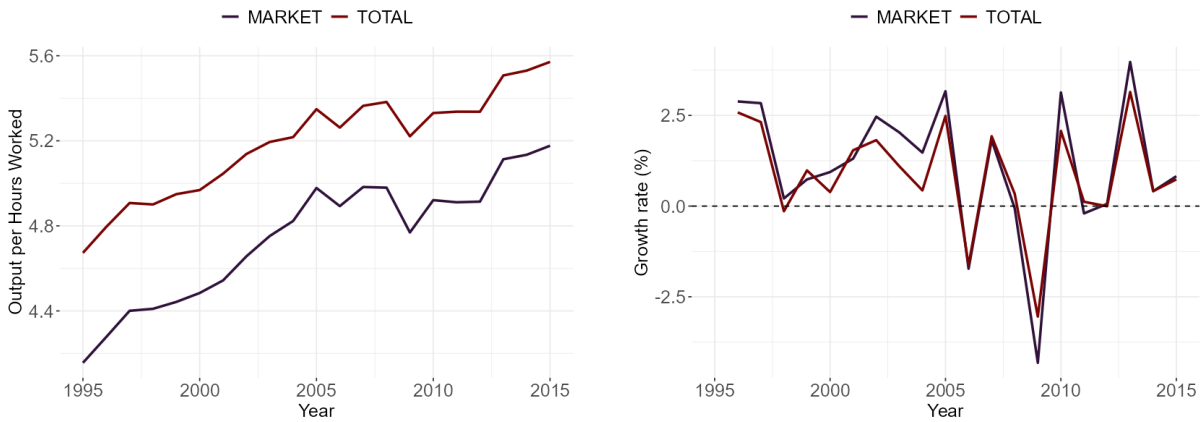
(a) Levels (b) Growth Rate

Figure 1: Output per Worker



(a) Levels (b) Growth Rate

Figure 2: Output per Hour Worked



(a) Levels (b) Growth Rate

Figure 3: Output per Hour Worked by Employees

Discussion: Broadly speaking, output per worker has followed a similar trend across all three measures. In levels, output per worker rises from 1995, with a slight dip around 1997–1998, before rising continuously until the global financial crisis in 2008–2009, which caused a significant drop, before recovering from 2010 onwards. In terms of growth rates, this dip is visible across all three figures, and a minor slowdown around 2006 is also present, though less pronounced in Figure 1.

As output per worker shows a similar trend across all three measures, we will continue to use *EMP* as the measure of labour, as it is more closely aligned with the theoretical growth accounting framework and yields contributions that are more straightforward to interpret.

Generally, it appears to make little difference whether we consider the total or market economy in Japan during this period. The only slight difference occurs in Figure 3, when measuring *output per worker with hours worked by employees*: looking at Figure 3a, the total economy shows a small increase in the growth rate around 1998–2000 that is not present in the market economy.

Economic context: The periods identified above can be linked to several major policy changes and institutional features that characterised Japan over 1995–2015, though we note these are offered as possible explanations rather than causal claims.

The slowdown in output per worker growth around 1997–1998 coincides with a period of significant financial stress in Japan. The April 1997 consumption tax hike and the Asian Financial Crisis weakened domestic demand and external trade, while a broad credit contraction - driven by non-performing loans accumulated during the asset bubble - depressed investment. Together, these factors likely weighed on output growth during this period, consistent with the slowdown visible across all three figures.

The slight divergence between the total and market economy around 2000 in Figure 3 may be related to the repeated fiscal stimulus packages implemented throughout the 1990s and early 2000s, which involved large public spending. Since the total economy includes public sector activity, this distinction may partly explain the divergence.

The sharp drop in 2008–2009 is consistent with the impact of the global financial crisis, which affected Japan's export-oriented economy through the collapse in global trade, particularly in manufacturing sectors such as automobiles and electronics.

Finally, the rebound from 2010 reflects a recovery from the global financial crisis. From 2012, Abenomics - combining aggressive monetary easing, flexible fiscal policy, and structural reform - is associated with a modest further improvement in growth. However, structural constraints, including an ageing workforce and weak domestic demand, continued to weigh on output per worker growth over the period.

C Physical and Human Capital

Decompose the growth in income per capita between capital, human capital, and TFP using the formula coming from the time derivative of the equation from class $y = (K/Y)^{\alpha/1-\alpha}hA$. Assume $\alpha = 1/3$. Discuss different measures of capital and human capital that you have available in the data and whether they make a difference for your growth accounting exercise.

Human Capital

Data choices: To capture human capital, we examine the following measures:

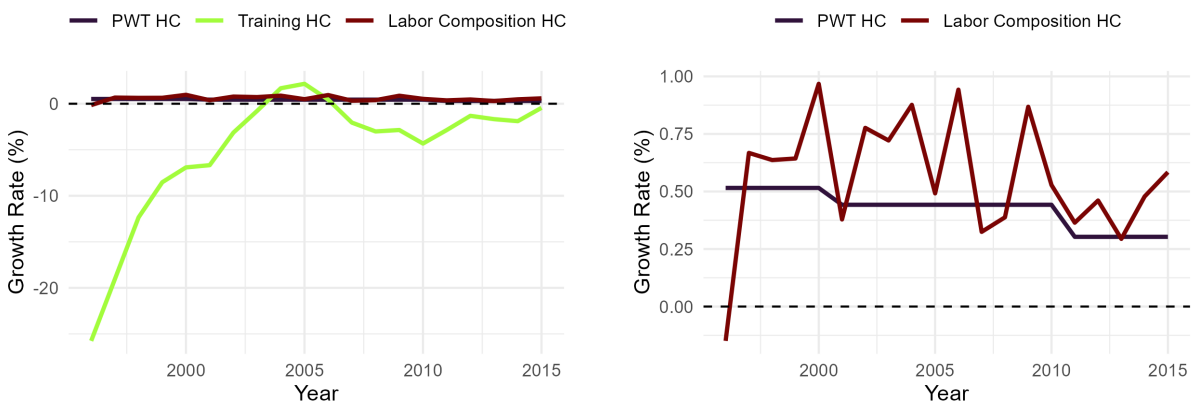
1. **Training Human Capital (HC)** [*Kq_Train*]: This represents investment in firm-specific human capital. It is measured as the sum of investment in vocational training and apprenticeships. We acknowledge that this measure does not entirely capture education and human capital, as it only represents on the job training.

2. **PWT Database HC**: The PWT database contains an aggregate human capital index (HC) based on the average years of schooling from Barro and Lee (2013)[1] and an assumed rate of return to education, based on Mincer equation estimates around the world. It is important to note that this variable is at the country level, and so does not give sector specific human capital indices.

3. **Labour Composition HC** [*LAB_QI / H_EMP*]: Following the definition of the EUKLEMS database, this measure divides a labour compensation index [real labour services, *LAB_QI*] by total hours worked by persons engaged [*H_EMP*]. It measures labour composition, which is seen to indicate quality. This variable is calculated using marginal productivities as its basis. This approach assumes that each worker's flow of labour services is proportional to the number of hours worked, and workers are paid their marginal productivity, for simplicity. We acknowledge that this variable can measure other things unrelated to human capital, such as the bargaining power of the workers coming from unionization or governments' minimum wage policies. This index has sectoral level heterogeneity.

Figures

Figure 4 plots the growth rates of the three human capital indices, while Figures 5, 6, and 7 present the resulting growth accounting decompositions - showing the contributions of TFP, capital, and human capital to output per worker growth for each of the three human capital measures, in annual and cumulative terms.

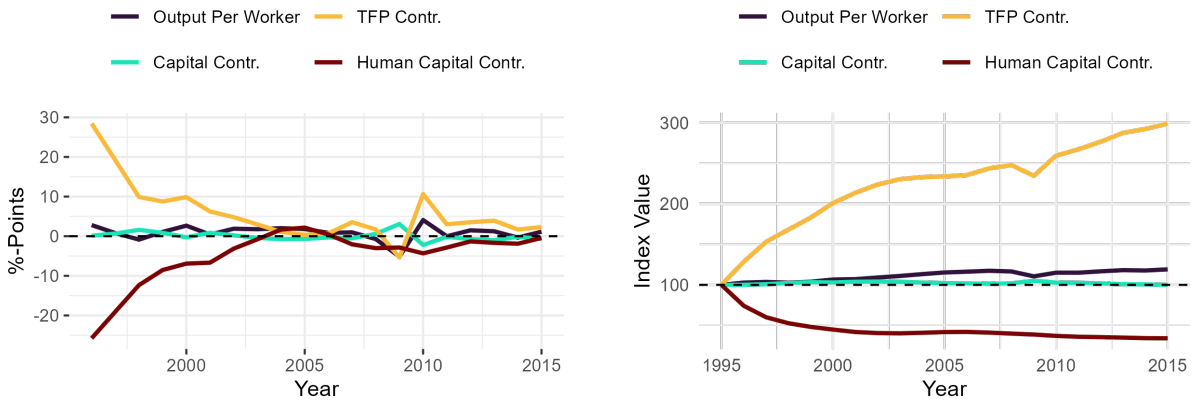


(a) With Training HC

(b) Without Training HC

Figure 4: Different Human Capital Indices Growth Rates (%)

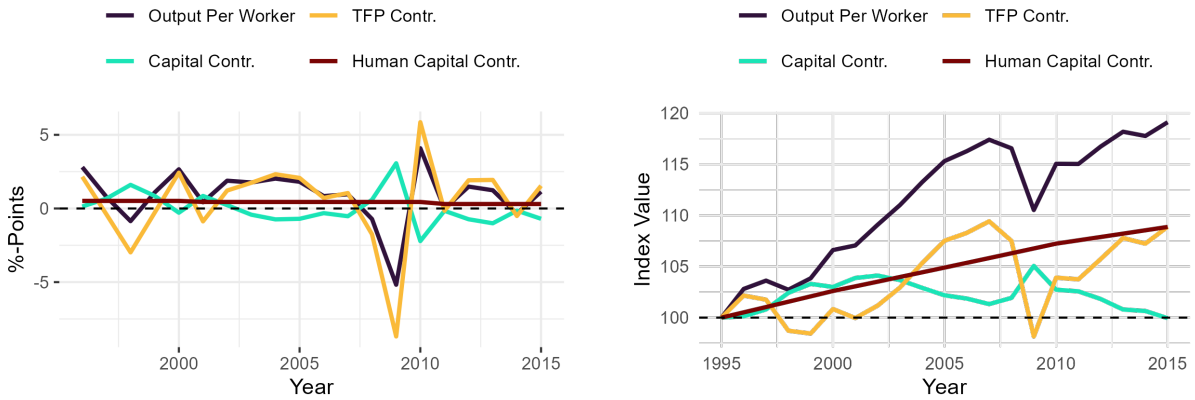
[1] Robert J. Barro and Jong Wha Lee (2013), "A new data set of educational attainment in the world, 1950–2010," Journal of Development Economics, 104, pp. 184–198. https://doi.org/10.1016/j.jdeveco.2012.10.001



(a) Each Year's Contributions (%-Points)

(b) Cumulative Contributions (1995=100)

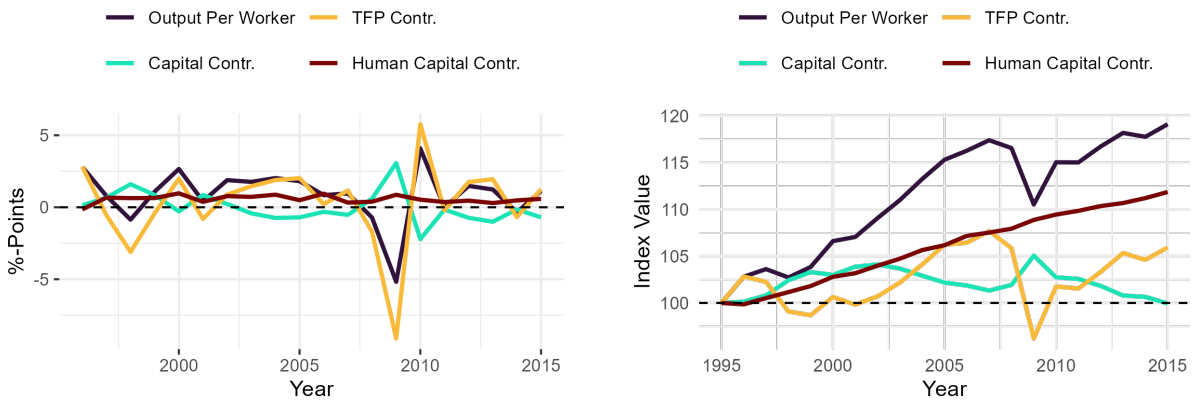
Figure 5: Output Per Worker Growth and Contributions (using Training HC)



(a) Each Year's Contributions (%-Points)

(b) Cumulative Contributions (1995=100)

Figure 6: Output Per Worker Growth and Contributions (using PWT HC)



(a) Each Year's Contributions (%-Points)

(b) Cumulative Contributions (1995=100)

Figure 7: Output Per Worker Growth and Contributions (using labour Comp. HC)

Discussion: Figure 4a examines the indices in terms of their growth rates. It reveals that the Training HC index is relatively volatile compared to what would be expected from a human capital index, likely because it captures only short-term firm-specific training expenditures rather than broader educational attainment, making it subject to year-to-year fluctuations in corporate training budgets. In order to examine the other two indices in detail, Figure 4b removes Training HC. This reveals that the two remaining indices track each other closely, with the PWT index following a notably smoother path, reflecting its basis in average years of schooling, which evolves gradually over time.

In order to assess what each human capital index implies for the growth accounting exercise, we run the decomposition with each measure separately. Figure 5 shows that using training HC creates very unlikely growth accounting results - around 1995, human capital's contribution to output per worker growth is approximately -20 percentage points, which appears implausible. While a negative contribution is not impossible in principle, a value of this magnitude is difficult to reconcile with the data. Figures 6 (PWT HC) and 7 (labour Comp. HC) produce more sensible results, broadly consistent results with one another. Figure 7 shows a higher cumulative improvement in human capital over the period, and also exhibits slightly more year-to-year variation in its contribution, perhaps reflecting the fact that labour composition can respond more quickly to changes in the workforce than a schooling-based index. As this variable also accounts for sectoral heterogeneity - unlike the PWT index, which is measured at the country level, the labour Comp. HC is more appropriate for this exercise, facilitating later sectoral analysis. Hence, we select the labour Composition HC index as our human capital measure.

Economic context: To relate these observed trends to Japan's economic landscape, two points are worth noting. Firstly, the sharp decline in the Training HC growth rate in the late 90s may partly reflect cuts in corporate training budgets during the banking crisis, rather than a genuine deterioration in human capital. Secondly, Japan is characterised by a highly educated workforce, making the large negative human capital contribution implied by the Training HC measure difficult to reconcile with the broader economic context.

Physical Capital

Data choices: To examine physical capital, we choose the following measures:

1. **Gross Fixed Capital Formation** [*Kq_GFCF*]: This is the real net capital stock, including all assets, and can be interpreted as capturing total K. This is the broadest measure we use for the physical capital stock.

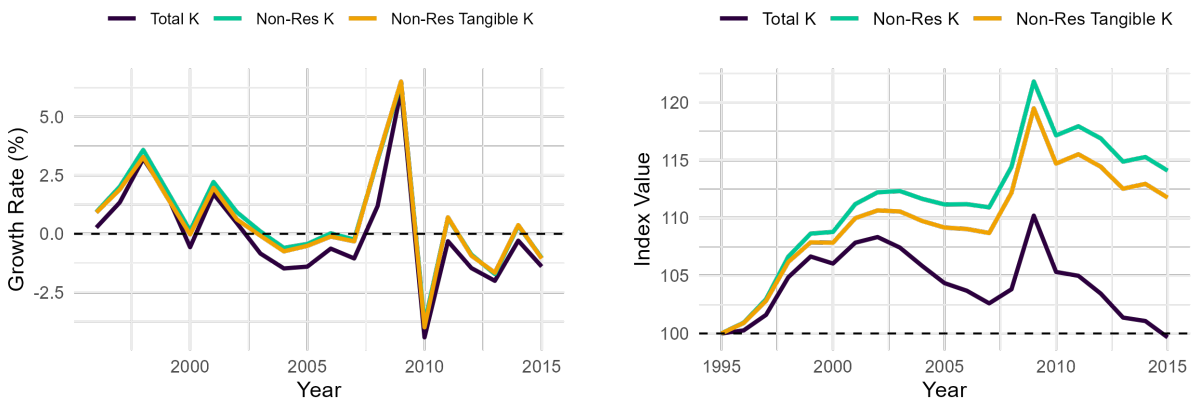
2. **Non_Residential Capital** [*NRes_K*]: This measure excludes residential structures from total capital [*Kq_GFCF*]. Residential capital mainly produces housing services for households, rather than market output, so removing it provides a measure of capital more closely related to productive economic activity. As this variable was not in the dataset for Japan, we computed it by deducting the capital stock of nominal residential structures [*K_Rstruc*] from the nominal total capital stock [*K_GFCF*], before deflating the result with the yearly price indices.

3. **Non_Residential Tangible Capital** [*NRes_Tang_K*]: This measure further excludes intangible assets such as R&D, software, and intellectual property products. It therefore captures only tangible, non-residential productive capital, providing a narrower measure of physical capital used in production. This variable was also not in the dataset, so we computed it by deducting the capital stock of nominal residential structures [*K_Rstruc*], R&D [*Kq_RD]*, software [*Kq_Soft_DB*], and in-

tellectual property [Kq_OIPP] from the total capital stock of nominal buildings [K_GFCF], then deflating the result with the yearly price indices.

Figures

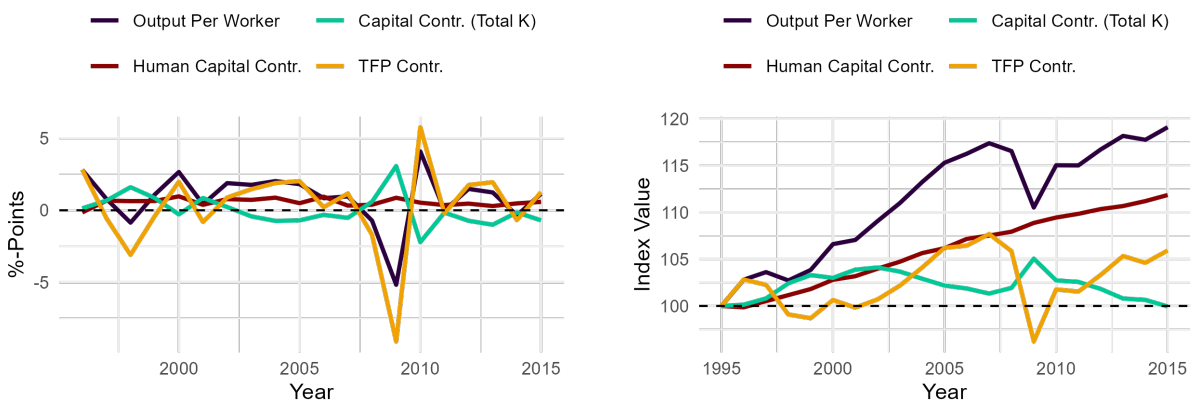
Figure 8 plots the growth rates and evolution of the three indices, while Figures 9, 10 and 11 present the resulting growth accounting decompositions for the each measure - showing the contributions of TFP, capital, and human capital to output per worker growth, both annually and cumulatively.



(a) Growth Rates (%)

(b) Index (1995=100)

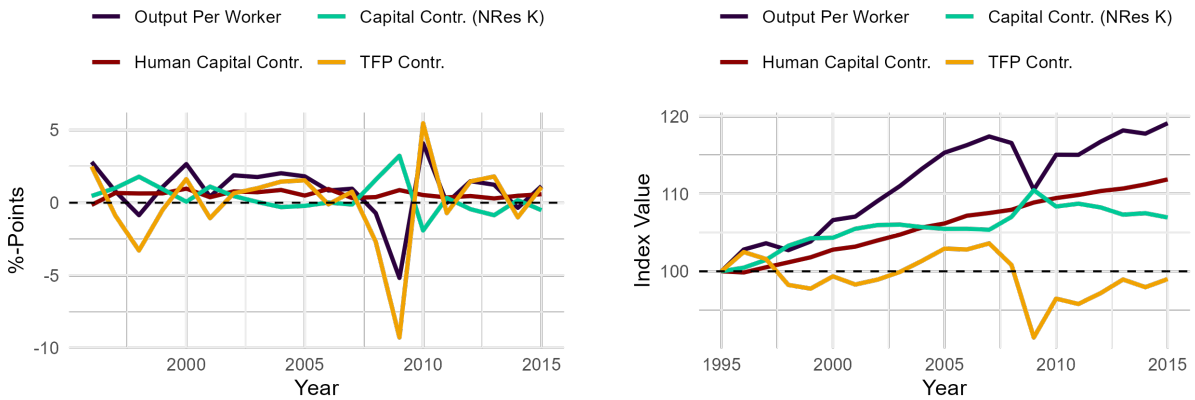
Figure 8: Different Physical Capital Indices



(a) Each Year's Contributions (%-Points)

(b) Cumulative Contributions (1995=100)

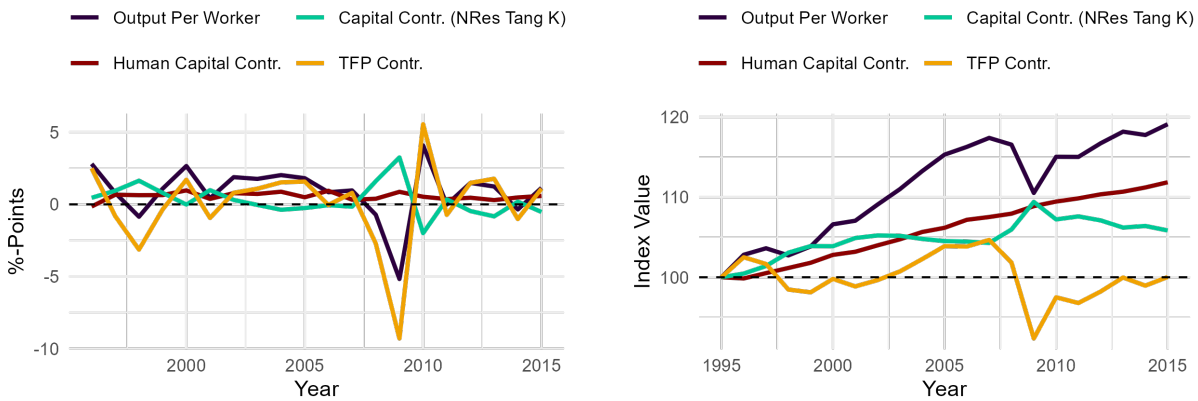
Figure 9: Output Per Worker Growth and Contributions (Using Total K)



(a) Each Year's Contributions (%-Points)

(b) Cumulative Contributions (1995=100)

Figure 10: Output Per Worker Growth and Contributions (Using Non-Residential K)



(a) Each Year's Contributions (%-Points)

(b) Cumulative Contributions (1995=100)

Figure 11: Output Per Worker Growth and Contributions (Using Non-Residential Tangible K)

Discussion: From Figure 8a, we see that the growth rates of the three indices are close to each other, following very similar patterns across the 20 year period. However, from Figure 8b, we see that the cumulative growth in total capital is lower than the other measures. Therefore, our first impression is that the choice of the physical capital stock measure may have some effects on our growth accounting results. In order to identify the impact of the choice of measure, we run our exercise with each of them separately.

From panel (a) of Figures 9, 10 and 11, the patterns of contributions appear very similar, as expected. In cumulative terms, results are similar when we use non-residential capital (Figure 10b) and non-residential tangible capital (Figure 11b). However, the cumulative graph in Figure 9b differs in terms of the contribution of physical capital. The contribution of physical capital to per capita growth is notably higher when we exclude residential structures, which is unsurprising given that capital measures excluding residential structures experience higher cumulative growth over the period, as shown in Figure 8b.

Residential structures are housing-services-related capital which, when we examine the data, is zero for all sectors except real estate activities (Sector L). Excluding them would therefore distort our sectoral results for Sector L - which will be explored in the subsequent section - so we retain them in our capital measure. Similarly, we choose not to exclude intangible assets, as they are important for software and

R&D intensive sectors such as information and communication activities and professional, scientific and technical services. Taking into consideration the sectoral importance of intangible assets and residential structures, the total capital stock [Kq_GFCF] is therefore the most appropriate measure.

In sum, for the rest of the growth accounting exercise, we use labour compensation and total capital stock to capture human and physical capital, respectively.

Economic context: Turning to Japan's broader economic context, the lower level of total capital relative to the non-residential measures, visible in Figure 8b, may partly reflect relatively weak residential capital accumulation following the collapse of the asset price bubble in the early 1990s, which contributed to slower growth of the overall capital stock over the subsequent decades. More broadly, capital growth rates across all measures remain relatively subdued, consistent with a persistently weak investment environment in Japan over the period.

D Sectoral Decomposition

Repeat your exercise by broad sectors of the economy (codes, A, B, C, D-E, F, G, H, I, J, K, L, M-N, and O-U). For which sector do you find the highest growth in output per worker? and the lowest? and the highest and lowest TFP growth?

Tables & Figures

In order to examine sectoral movements, Table 1 reports the average output per worker and TFP growth rates by sector over the period, while Figures 12 and 13 plot the output per worker and TFP indices by sector over the full period (1995=100).

NACE Code	OPW Growth	Output Share
C	2.9	19.7
J	2.6	4.4
A	1.8	1.7
K	1.4	4.3
M-N	1.1	6.9
G	0.3	16.0
H	0.1	6.2
L	-0.4	11.7
D-E	-0.6	4.1
F	-0.8	6.9
O-U	-0.9	14.5
I	-2.0	3.5
B	-2.6	0.2

(a) Output Per Worker Growth

NACE Code	TFP Growth	Output Share
C	2.5	19.7
A	2.2	1.7
J	1.8	4.4
M-N	1.2	6.9
L	-0.2	11.7
G	-0.6	16.0
H	-0.7	6.2
K	-0.8	4.3
D-E	-1.2	4.1
F	-1.9	6.9
I	-2.8	3.5
O-U	-2.9	14.5
B	-4.5	0.2

(b) TFP Growth

Table 1: Average Output Per Worker (OPW) and TFP Growth by Sector (Japan, 1996–2015)

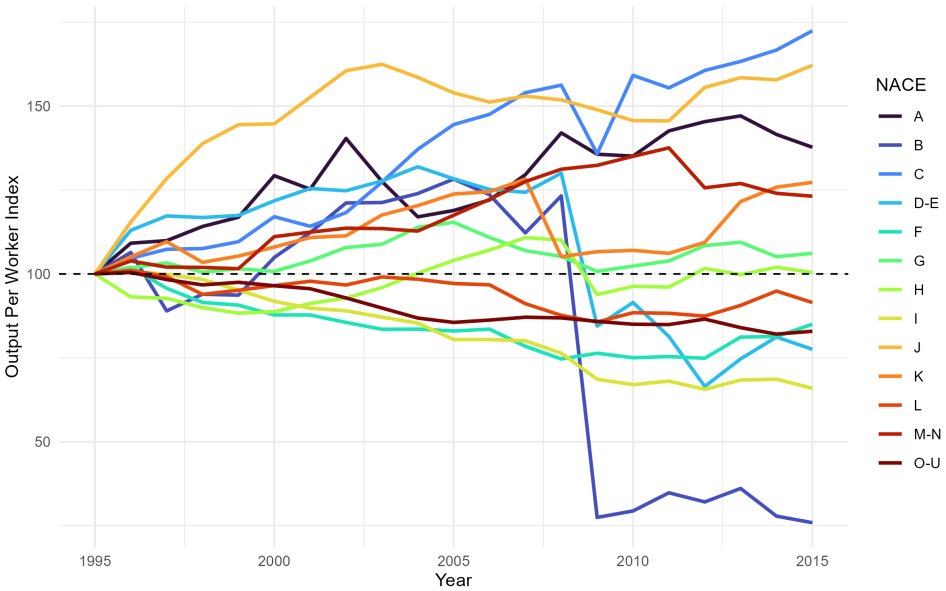


Figure 12: Output Per Worker Index by Sector (1995=100)

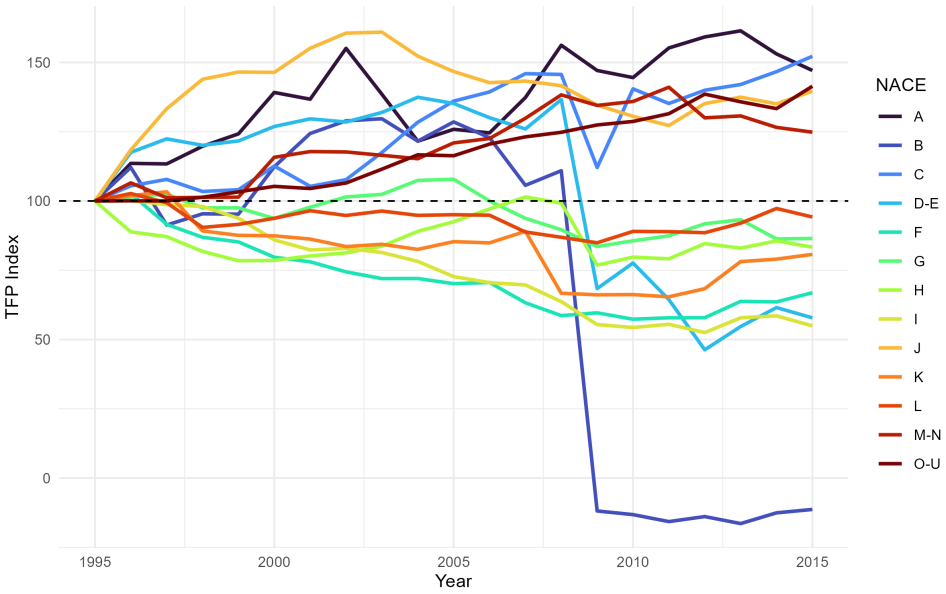


Figure 13: TFP Index by Sector (1995=100)

Growth in Output Per Worker

Discussion: According to Table 1a, the highest average output per worker growth over the period is observed in the manufacturing sector (C) at 2.9%, which is also the largest sector by output share at 19.7%. It is followed by the information and communication sector (J) at 2.6%. On the other hand, the mining and quarrying sector (B) records the lowest average output per worker growth at -2.6%.

Figure 12 reinforces these findings and reveals further patterns. There is considerable dispersion across sectors, with several sectors falling below the 1995 baseline by 2015, indicating negative output per worker growth over the period. The 2008–2009 global financial crisis is clearly visible across nearly all sectors, with the mining and quarrying sector (B) experiencing a particularly sharp drop. The information and communication sector (J) was the fastest growing between 1995–2006, before plateauing in the latter part of the period. The transport and storage sector (I) exhibits a sustained and steady decline throughout the entire period.

Economic context: In terms of Japan's economic context, a few possible explanations are worth noting. The strong performance of manufacturing (C) is perhaps surprising, given the yen appreciation and offshoring pressures of the period, possibly suggesting genuine productivity gains within the sector. The plateauing of information and communication (J) after 2006 may reflect the maturation of the ICT sector, following the rapid expansion of the early 2000s. The sustained decline in transport and storage (I) could partly reflect weak domestic demand and demographic pressures associated with Japan's ageing population, though we cannot establish this causally.

Growth in TFP

Description: According to Table 1b, the highest average TFP growth is observed in the manufacturing sector (C) at 2.5%. This is followed by agriculture, forestry and fishing (A) at 2.2% and information and communication (J) at 1.8%, growing particularly rapidly in level terms between 1995 and 2006 before plateauing, as visible in Figure 13. The lowest TFP growth is recorded in mining and quarrying (B) at -4.5%, which experiences a particularly sharp collapse in levels around 2008 before partially recovering.

The sectoral rankings by TFP growth are broadly similar to those by output per worker growth, and while Figure 13 shows some dispersion, sectors move broadly similarly in direction over the period. Notably, since TFP is measured as a residual, strong conclusions about sectoral productivity differences are difficult to draw - sectors with high output growth that cannot be attributed to capital or labour will mechanically record higher TFP, limiting the interpretability of any differences.

Economic context: A few possible explanations for the sectoral TFP patterns are notable. The sharp collapse of mining and quarrying (B) may partly reflect the sector's exposure to the global collapse in commodity demand during the financial crisis. The strong performance of agriculture (A) is consistent with well-documented productivity improvements in this industry.

E Labour Shares by Sectors

Compute the labour share for each of these broad sectors (averaged across years). This object corresponds to $(1-\alpha)$ in our model. Repeat the previous exercise using the implied sector-specific α rather than assuming it is 1/3. Do the conclusions from your previous exercise change substantially?

Data choices: We consider two alternative measures of labour compensation for calculating labour share. First is using compensation of employees [$COMP$], and the second one is compensation of employees and self-employed [LAB], as is suggested by the EUKLEMS documentation, using the formula:

$$LAB = COMP + \frac{COMP}{HEMPE}(HEMP - HEMPE)$$

In order to check these alternatives, we compare the level of labour shares they imply for the whole country, averaged over 1995–2015. The first indicator [$COMP$] implies a labour share of 0.54, while the second indicator [LAB] implies a labour share of 0.63. These values correspond to capital share values of $\alpha = 0.46$ and $\alpha = 0.37$, respectively. As a general approximation, we take $\alpha \approx 0.33$, which was also assumed at the start of this project. The LAB-based estimate, $\alpha = 0.37$, is closer to this benchmark. Since it is closer to the generally accepted level and also includes the compensations paid to the self-employed, we select LAB indicator for the calculation of our labour shares.[2]

Tables & Figures

Table 2 reports the average labour and capital shares by sector over 1995–2015. Table 3 compares average TFP growth rates using the country-level $\alpha = \frac{1}{3}$ against sector-specific α values. Figure 14 plots the resulting TFP index by sector (1995=100) using the sector-specific α.

NACE Code	Labor Share	Capital Share
L	0.11	0.89
D-E	0.31	0.69
A	0.34	0.66
K	0.44	0.56
J	0.44	0.56
C	0.55	0.45
B	0.63	0.37
G	0.66	0.34
H	0.71	0.29
M-N	0.72	0.28
O-U	0.80	0.20
I	0.82	0.18
F	0.90	0.10

Table 2: Average labour and Capital Shares by Sector (1995–2015)

[2] If the labour share calculated for a sector using this method gives a labour share of higher than one, we turn back to the simpler COMP indicator for calculating the labour share for that particular sector.

NACE Code	TFP Growth	Output Share
C	2.5	19.7
A	2.2	1.7
J	1.8	4.4
M-N	1.2	6.9
L	-0.2	11.7
G	-0.6	16.0
H	-0.7	6.2
K	-0.8	4.3
D-E	-1.2	4.1
F	-1.9	6.9
I	-2.8	3.5
O-U	-2.9	14.5
B	-4.5	0.2

(a) Country Level α (1/3)

NACE Code	TFP Growth	Output Share
L	7.7	11.7
A	6.0	1.7
C	2.6	19.7
J	2.1	4.4
O-U	1.7	14.5
M-N	1.1	6.9
H	-0.6	6.2
G	-0.6	16.0
F	-1.4	6.9
I	-2.5	3.5
K	-3.3	4.3
D-E	-3.8	4.1
B	-4.8	0.2

(b) Sector-Specific α

Table 3: Average TFP Growth by Sector (Japan, 1996–2015)

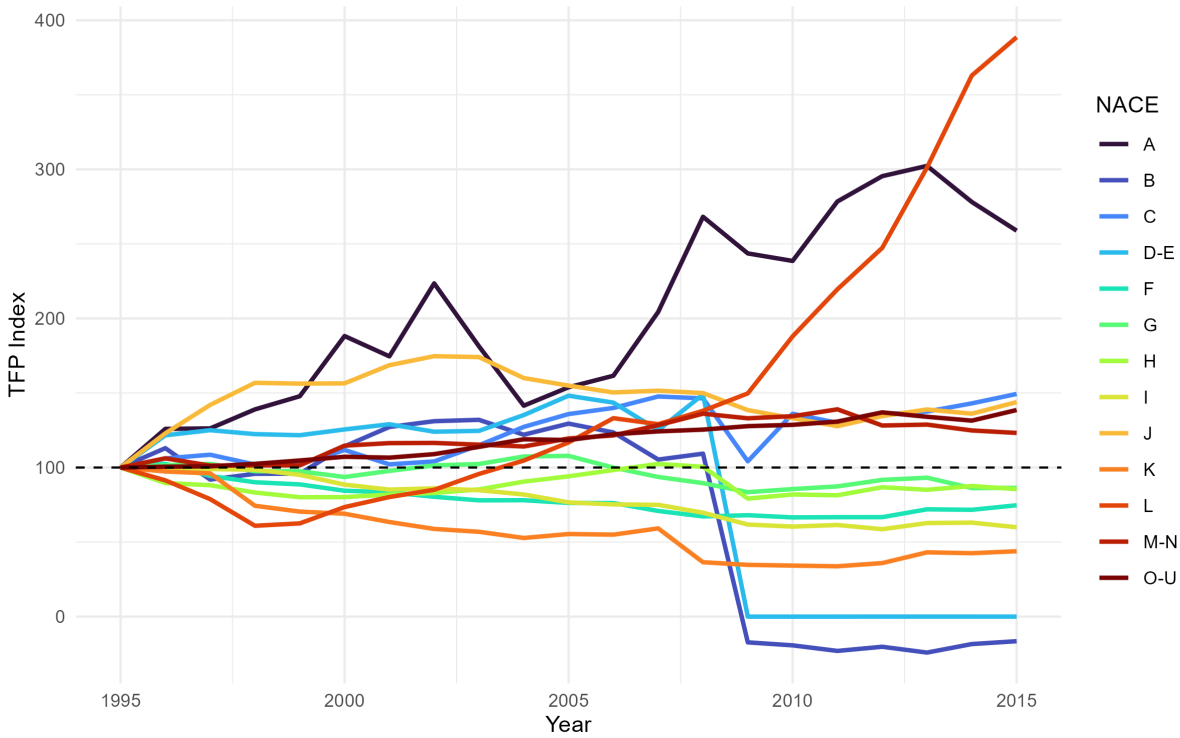


Figure 14: TFP Index by Sector (1995=100)

Description: Table 2 reveals considerable variation in labour and capital shares across sectors. Notably, Sector L (real estate services) has a capital share of 0.89, far above the assumed country-level $\alpha = \frac{1}{3}$, while Sector F (Construction) has a labour share of 0.90, implying a capital share of just 0.10. This dispersion motivates the use of sector-specific α values.

Table 3 demonstrates that switching to sector-specific α values does change some results, though broad conclusions are largely preserved. The top-performing sectors C (Manufacturing), A (Agriculture, Forestry and Fishing), and J (Information and Communication) remain among the highest TFP growth sectors, and Sector B (Mining and Quarrying) remains the lowest at -4.8%, consistent with the previous results. The most notable change is sector L, which jumps from -0.2 to 7.7, as expected given its capital share deviates most from the country-level assumption. Several other sectors also see meaningful changes, suggesting that for sectors whose sector-specific α differs substantially from the country-level assumption, the choice of α matters for the decomposition.

Figure 14 reinforces these findings. Sector L picks up strongly after the 2008–2009 crisis, becoming the highest TFP sector by 2015[3], while Agriculture (A) retains its strong performance but shows more volatility in its TFP path compared to the country-level α results. Overall, while the broad conclusions from the previous exercise are largely preserved, sector-specific α values produce meaningful differences for capital-intensive or labour-intensive sectors, and should be preferred where sectoral heterogeneity in factor shares is significant.

F Aggregation

How would you compute aggregate growth in income per worker, and capital using the sectoral data? Does it coincide with the reported aggregates?

Data choices: We use the weighted average of each sectors' growth in income per worker, TFP and capital using sectoral value-added shares as weights.

Tables & Figures

Table 4 compares the growth rates of income per worker, TFP, and capital derived from sectoral aggregation compared to total economy data over the period, while Figure 15 presents these two series for each of the three variables.

[3] We had a closer look at Sector L by decomposing its contributions in Appendix Figure A1. This shows that capital has significantly fallen from 2000 onwards, while output per worker remained stable. This drop has been absorbed by the TFP residual, leading to a sharp increase from 2009 onwards. For a more detailed explanation, please refer to the appendix.

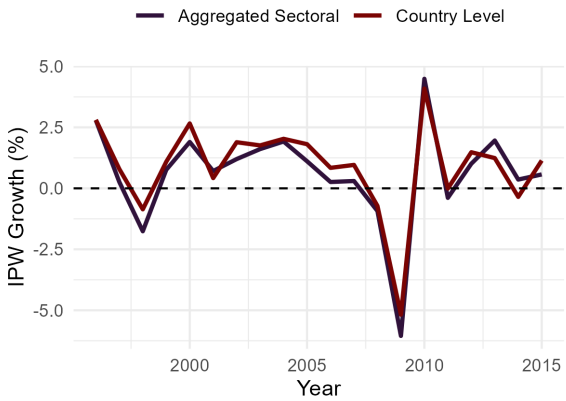
Year	Income per Worker	TFP Growth	Capital Growth
1996	2.8	2.4	3.4
1997	0.3	-1.5	2.9
1998	-1.8	-5.6	2.0
1999	0.8	0.0	1.2
2000	1.9	4.2	1.6
2001	0.7	0.1	1.3
2002	1.2	2.1	0.8
2003	1.6	3.6	0.7
2004	1.9	4.2	0.9
2005	1.1	3.5	1.0
2006	0.3	1.6	1.5
2007	0.3	0.6	1.2
2008	-0.9	-0.1	0.3
2009	-6.0	-11.4	-0.6
2010	4.5	11.1	-0.7
2011	-0.4	0.7	-0.4
2012	1.0	2.5	-0.1
2013	2.0	5.5	0.3
2014	0.4	2.4	0.5
2015	0.6	2.5	0.8

(a) Sectoral Aggregation

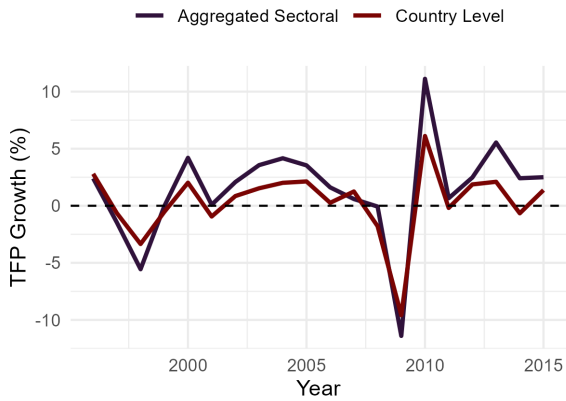
Year	Income per Worker	TFP Growth	Capital Growth
1996	2.8	2.8	3.4
1997	0.8	-0.7	2.8
1998	-0.9	-3.4	2.1
1999	1.1	-0.5	1.8
2000	2.7	2.0	1.7
2001	0.4	-0.9	1.4
2002	1.9	0.8	0.9
2003	1.8	1.5	0.7
2004	2.0	2.0	0.7
2005	1.8	2.1	0.8
2006	0.8	0.3	0.8
2007	1.0	1.2	0.5
2008	-0.7	-1.8	0.1
2009	-5.2	-9.6	-0.6
2010	4.1	6.1	-0.7
2011	0.0	-0.2	-0.5
2012	1.5	1.9	-0.2
2013	1.2	2.1	0.0
2014	-0.3	-0.7	0.2
2015	1.1	1.4	0.3

(b) Total Economy

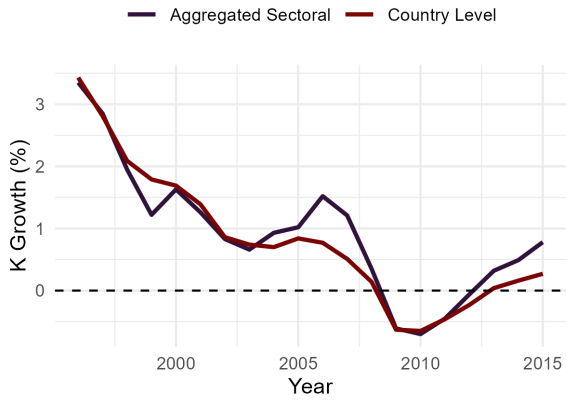
Table 4: Growth Rates from Sectoral Aggregations versus Total Economy Data



(a) Income Per Worker Growth



(b) TFP Growth



(c) Capital Growth

Figure 15: Growth Rates from Sectoral Aggregations versus Total Economy Data

Description: As shown in Table 4, the sectoral aggregation largely coincides with the total economy data, with correlations of 0.97, 0.95 and 0.96 for income per worker, TFP and capital growth respectively. The small remaining differences are expected, as aggregation using value-added weights introduces compositional differences. Figure 15 confirms that the aggregation successfully approximates the country-level series for all three variables, though the gap is most visible in capital growth, where the two series diverge more noticeably in the earlier part of the period.

G Policy & Economic Changes

Relate any major policy changes or state of the economy to the evolution of the aggregate or sectoral time series that you have analysed for your country and time of study.

Summary of results: This growth accounting exercise has documented several key patterns in Japan's economic evolution over 1995–2015. Aggregate output per worker grew modestly, with a dip around 1997–1998 and a sharp drop in 2008–2009, before a partial recovery. Decomposing this growth using our preferred measures, human capital emerges as the largest contributor, followed by TFP, while physical capital's contribution remained subdued throughout. At the sectoral level, considerable heterogeneity is evident - manufacturing (C), agriculture (A), and information and communication (J) were the strongest performers in both output per worker and TFP growth, while mining (B), transport (I), and several service sectors experienced sustained declines. These broad conclusions are largely preserved when using sector-specific α values, as discussed in Section E.

Economic context: Throughout this exercise, we have noted possible links between the observed trends and Japan's economic landscape. Here we draw these together more explicitly, though we emphasise that growth accounting cannot establish causality and these are offered as possible explanations for the trends we observe. The subdued capital contribution is consistent with the weak investment environment that characterised Japan following the asset price bubble collapse, during which declining profitability expectations and the misallocation of credit to unproductive zombie firms are argued to have depressed investment. The dip in aggregate output per worker around 1997–1998 coincides with the consumption tax hike, the Asian Financial Crisis, and the banking crisis, all of which weakened output growth during this period. The sharp drop in 2008–2009 may reflect Japan's export-oriented economy's exposure to the global collapse in trade amid the global financial crisis.

At the sectoral level, the strong performance of manufacturing, despite yen appreciation and offshoring pressures, may suggest genuine productivity gains, while the decline in transport and several service sectors is consistent with weak domestic demand and demographic pressures. The zombie firm literature offers a plausible explanation for the broader pattern of weak productivity growth across several sectors. Finally, as noted in Section B, the recovery from 2010 likely reflects a post global financial crisis rebound, with Abenomics from 2012 associated with a modest further improvement in growth, though persistent structural constraints - including demographic pressures and sluggish productivity growth - continued to weigh on growth throughout the period.

Conclusion: Overall, Japan's experience over 1995–2015 illustrates how financial crises, structural rigidities, and demographic pressures can shape the pace and composition of growth. As reiterated throughout, while growth accounting provides a useful framework for decomposing these patterns, it remains descriptive in nature and cannot establish the causal mechanisms underlying them.

Appendix

In this appendix, we aim to give more context for the sharp TFP increase in the Real-Estate Sector. As noted, starting around 2000, the capital contribution to growth in Sector K significantly drops while output per worker remains relatively stable. This drop is therefore absorbed by TFP, leading to an increase of its contribution.

Economic context: The decline in the new housing construction after 1990s lowers capital accumulation in this sector. This decline is mainly caused by the asset price bubble collapse that happened in 1990s which especially hit the inflated real estate and stock market. The decreasing capital in this sector is further strengthened by the lower levels of demand caused by the declining population growth rates. Japan's population growth had a decreasing pattern in this period, and it even started to go into negative territories after 2009, decreasing the total population (FRED, 2023 & FRED, 2026)[4]. The trend of capital growth in the sector K is especially affected by these developments.

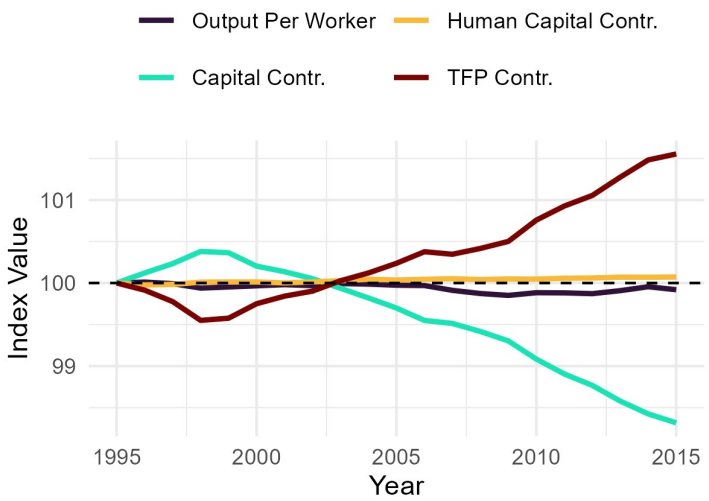


Figure A1: Output Per Worker Growth and Cumulative Contributions (L, Real-Estate Sector)

[4]FRED. (2023). Work Started: Construction: Dwellings and Residential Buildings: Total for Japan. Available at https://fred.stlouisfed.org/series/WSCNDW01JPA489N. FRED. (2026) World Bank, Population Growth for Japan [SPPOPGROWJPN], retrieved from FRED, Federal Reserve Bank of St. Louis; https://fred.stlouisfed.org/series/SPPOPGROWJPN, March 8, 2026.